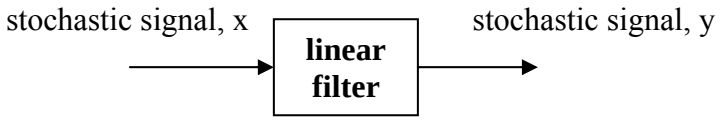
Linear filtering of stochastic signals

Overview of the most important concepts, results and applications

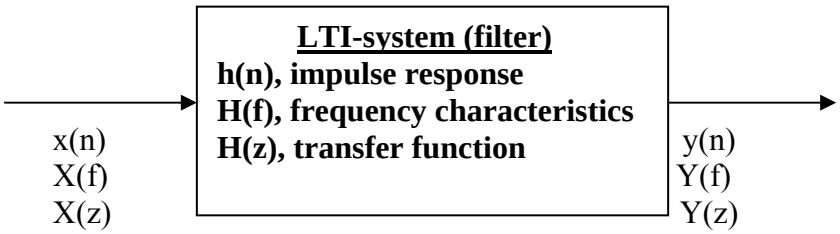
Problem



System-classification

- "lumped" system (can be described with a finite number of differential equations)
- linear system (super-posistion applies)
- time-variant system (system characteristics are constant with the time)
- causal system (system-output only depends of previous inputs and calculated output)
- LTI-system = linear + time-invariant (only this is necessary for the following results)

System theory for LTI-systems and deterministic or stochastic signals (repetition)



$$y(n) = h(n) * x(n) = \sum_k h(k)x(n-k)$$ discrete convolution

$$Y(f) = H(f)X(f)$$ DTFT, discrete-time Fourier-transf.

$$Y(z) = H(z)X(z)$$ Z-transformed

Filtering of discrete-time stochastic signals (main results)
SEE NEXT PAGE FOR AN OVERVIEW !!!

Continuous-time (analogue) signals and systems

The same results apply as for discrete-time (sampled) signals, but

- replace discrete convolution with continuous convolution (replace total with integral and k with τ)
- replace impulse response h(n) with h(t)
- replace DTFT (discrete-time Fourier transf.) with CTFT (continuous-time FT)
- replace $R_{xx}(k)$ with $R_{xx}(\tau)$
- replace the Z-transformation with the Laplace-transformation

Typical examples of application

- Filtering of white noise to band limited white noise, noise equivalent band with
- Pulse-shaping of digital symbols for analogue transmission signal, power spectrum
- System-identification of unknown system, e.g. based upon formula (5) or (9) on next page

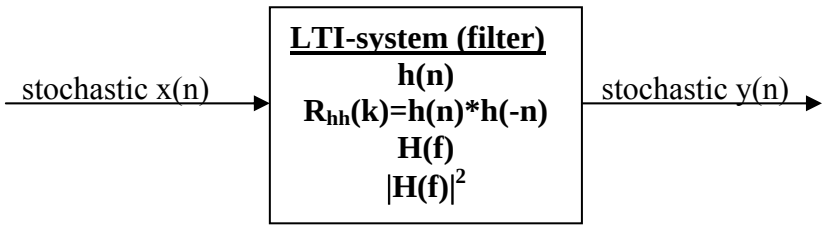
Filtering of discrete-time stochastic signals (main results)

Linear filtering does not change the stationarity characteristics of a stochastic signal, i.e. e.g.

- x(n) is SSS (strict-sense stationary) → y(n) is SSS
- x(n) is WSS (wide-sense stationary) → y(n) is WSS

Input-Output relations

discrete, deterministic LTI-system



$x(n)$	(stoch. signal)	$y(n) = h(n) * x(n)$	(1a)
$m_x = E[x(n)]$	(mean value)	$m_y = m_x \cdot H(0)$	(2a)
$\sigma_x^2 = E[(x(n) - m_x)^2]$	(power)	$\sigma_y^2 = R_{yy}(0)$	(3a)
$R_{xx}(k) = E[x(n)x(n+k)]$	(autocorr.fct.)	$R_{yy}(k) = R_{hh}(k) * R_{xx}(k)$	(4a)
$S_{xx}(f) = DTFT\{R_{xx}(k)\}$	(power spectrum)	$S_{yy}(f) = \lvert H(f)\rvert^2 S_{xx}(f)$	(5a)
	(crosscorr. fct)	$R_{xy}(k) = E[x(n+k)y(n)]$	
		$= h(-k) * R_{xx}(k)$	(6a)
		$R_{yx}(k) = E[x(n)y(n+k)]$	
		$= h(k) * R_{xx}(k)$	(7a)
	(cross spectrum)	$S_{xy}(f) = H^*(f) S_{xx}(f)$	(8a)
		$S_{yx}(f) = H(f) S_{xx}(f)$	(9a)
$S_{xx}(z) = Z\{R_{xx}(k)\}$	(Z-transformed)	$S_{yy}(z) = \lvert H(z)\rvert^2 S_{xx}(z)$	
		$= H(z)H(z^{-1})S_{xx}(z)$	(10a)
		$S_{xy}(z) = H(z^{-1})S_{xx}(z)$	(11a)
		$S_{yx}(z) = H(z)S_{xx}(z)$	(12a)

Is especially **x(n) (aproximated) gauss-distributed white noise, then**

$x(n) = N(0, \sigma^2)$	(stoch. signal)	$y(n) = h(n) * x(n)$	(1b)
$m_x = 0$	(mean value)	$m_y = 0$	(2b)
$\sigma_x^2 = \sigma^2$	(power)	$\sigma_y^2 = R_{yy}(0)$	(3b)
$R_{xx}(k) = \sigma^2 \cdot \delta(k)$	(autocorr.fct.)	$R_{yy}(k) = \sigma^2 \cdot R_{hh}(k)$	(4b)
$S_{xx}(f) = \sigma^2$	(power spectrum)	$S_{yy}(f) = \sigma^2 \cdot \lvert H(f)\rvert^2$	(5b)
	(crosscorr. fct.)	$R_{xy}(k) = \sigma^2 \cdot h(-k)$	(6b)
		$R_{yx}(k) = \sigma^2 \cdot h(k)$	(7b)
	(cross spectrum)	$S_{xy}(f) = \sigma^2 \cdot H^*(f)$	(8b)
		$S_{yx}(f) = \sigma^2 \cdot H(f)$	(9b)